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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 1 of 10 (Start time: 07:16:29 AM, 26 June 2025) **Total Marks:** 1

Which term is often chosen to reparameterize the proper time ?

Select the correct option

Reload Math Equations

- $\tau = \frac{s}{c}$
- $\tau = s \times c$
- $\tau = \frac{s}{c^2}$
- $\tau = \frac{c}{s}$

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Quiz

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 2 of 10 (Start time: 07:17:33 AM, 26 June 2025) **Total Marks:** 1

The absolute derivative of T^i along the curve $C : x^i = x^i(t)$ is denoted by _____.

Select the correct option

Reload Math Equations

- $\frac{\delta T^i}{\partial t}$
- $\frac{dT^i}{dt}$
- $\frac{\delta T^i}{\delta t}$ MahrooChudhary
- dT

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 3 of 10 (Start time: 07:18:36 AM, 26 June 2025) **Total Marks:** 1

Which of the following is the correct mathematical form of the Bianchi identity?

Select the correct option

Reload Math Equations

- $\nabla_k R_{ij} - \nabla_i R_{jk} + \nabla_j R_{ki} = 0$
- $R^i_{jkl} = -R^i_{jkl}$
- $R_{ijkl} = -R_{jikl}$
- (selected) $R_{ijkl} + R_{ikdj} + R_{iljk} = 0$

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Quiz

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 4 of 10 (Start time: 07:19:41 AM, 26 June 2025) **Total Marks: 1**

What is the expression for the relativistic mass $\hat{m}$ if m is rest mass of the particle?

Select the correct option

Reload Math Equations

- $m = \frac{\hat{m}}{\sqrt{1 - \frac{\hat{v}^2}{c^2}}}$
- $\hat{m} = \frac{mc}{\sqrt{1 - \frac{\hat{v}^2}{c^2}}}$
- (selected) $\hat{m} = \frac{m}{\sqrt{1 - \frac{\hat{v}^2}{c^2}}}$
- $\hat{m} = \frac{mv}{\sqrt{1 - \frac{\hat{v}^2}{c^2}}}$

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 5 of 10 (Start time: 07:21:04 AM, 26 June 2025) **Total Marks:** 1

How Riemann tensor of second kind is converted into Riemann tensor of first kind?

Select the correct option

Reload Math Equations

- ○ Raising an index with the metric tensor.
- ○ Raising two index with the metric tensor.
- ○ Lowering an index with the metric tensor. ✓
- ◉ Lowering two index with the metric tensor.

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 5 of 10 (Start time: 07:21:04 AM, 26 June 2025) **Total Marks:** 1

How Riemann tensor of second kind is converted into Riemann tensor of first kind?

Select the correct option

Reload Math Equations

- ○ Raising an index with the metric tensor.
- ○ Raising two index with the metric tensor.
- ◉ Lowering an index with the metric tensor.
- ○ Lowering two index with the metric tensor.

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 6 of 10 (Start time: 07:22:19 AM, 26 June 2025) **Total Marks:** 1

Minkowski spacetime is defined as:

Select the correct option

Reload Math Equations

- $g_{\mu\nu} = (dx^0)^2 - \Sigma_{i=1}^{3}(dx^i)^2$
- $g_{\mu\nu} = (dx^0)^2 - (dx^i)^2$
- $g_{\mu\nu} = (dx^0)^2 - \Sigma_{i=0}^{3}(dx^i)^2$
- $g_{\mu\nu} = \Sigma_{i=0}^{3}(dx^i)^2$

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 7 of 10 (Start time: 07:23:42 AM, 26 June 2025) **Total Marks:** 1

What is true for unit sphere?

Select the correct option

Reload Math Equations

- r = 2
- r = 1
- r = 0
- r = -1

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 8 of 10 (Start time: 07:24:33 AM, 26 June 2025) **Total Marks:** 1

What is a four vector in the special relativity?

Select the correct option

Reload Math Equations

- ○ A vector that only exists in non-relativistic physics.
- ◉ A vector with three spatial components and one time component.
- ○ A vector in three-dimensional space.
- ○ A vector with four spatial components.

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 9 of 10 (Start time: 07:25:20 AM, 26 June 2025)

Total Marks: 1

The basic rules for differentiation from Calculus carry over to covariant differentiation but do not fully apply to absolute differentiation.

Select the correct option

Reload Math Equations

- (●) false
- ( ) true

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MTH623 - Tensor Analysis and its Applications (Quiz # 4)

Quiz Start Time: 07:16 AM, 26 June 2025

Question # 10 of 10 (Start time: 07:26:06 AM, 26 June 2025) **Total Marks:** 1

Under which operation the set of Lorentz matrices form a group called the Lorentz group?

Select the correct option

Reload Math Equations

- ○ All of these
- ○ Subtraction of Matrices
- ◉ Multiplication of Matrices
- ○ Addition of Matrices

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